

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409930
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Stroud; Douglas Dwayne

Safety and stability device for an aircraft

Abstract

Systems and methods for a gyroscopic rotational wing for an aircraft are disclosed. In one embodiment, a safety and stability device for an aircraft comprises an inner ring, an outer ring that rotates relative to the inner ring, and a motor connected to the inner ring that drives rotation of the outer ring relative to the inner ring. In some embodiments, the safety and stability device rotates in a substantially horizontal plane and at a rotational speed sufficient to provide gyroscopic stability for the aircraft.

Inventors:	Stroud; Douglas Dwayne (West Plains, MO)
Applicant:	AeroLuxury LLC (West Plains, MO)
Family ID:	89384082
Assignee:	AeroLuxury LLC (West Plains, MO)
Appl. No.:	18/506466
Filed:	November 10, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240286739 A1	Aug. 29, 2024

Related U.S. Application Data

continuation parent-doc US 17815483 20220727 US 11851161 child-doc US 18506466

Publication Classification

Int. Cl.: **B64C17/06 (20060101)**

U.S. Cl.:

CPC **B64C17/06** (20130101);

Field of Classification Search

CPC: B64C (17/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0082421	12/2004	Perlo et al.	N/A	N/A
2010/0051754	12/2009	Davidson	N/A	N/A
2010/0320333	12/2009	Martin	N/A	N/A
2016/0362178	12/2015	Nam	N/A	N/A
2018/0178906	12/2017	Marcel	N/A	N/A
2020/0290726	12/2019	Sato et al.	N/A	N/A
2020/0391848	12/2019	Shi	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
113086216	12/2020	CN	N/A
113613996	12/2020	CN	N/A
06024391	12/1993	JP	N/A
2020005099	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for PCT/US2023/070942, dated Nov. 10, 2023.
pp. 12. cited by applicant
Chinese Patent Office, Office Action regarding Chinese Application No. 202380025557.8, mailed
Dec. 2, 2024. cited by applicant

Primary Examiner: Davis; Richard G
Attorney, Agent or Firm: Miller Johnson

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/815,483 filed Jul. 27, 2022, which is incorporated by reference herein in its entirety.

FIELD

(1) The present invention relates generally to flying aircrafts. More particularly, the present invention relates to systems and methods for using a spinning gyroscopic wing for aviation purposes.

BACKGROUND

(2) Air travel is a common mode of transportation. However, commercial air travel can be uncomfortable, expensive, and inconvenient given the rigorous regulation on the airline industry

and the relatively limited availability of flights between some locations. Additionally, aircrafts are operated by a limited number of people who qualify as licensed pilots because the learning curve to fly an aircraft is extremely steep, for example, steeper than driving an automobile. Still, air travel is an effective mode of travel because air travel typically represents the shortest path between Point A and Point B and because air travel can safely occur at much higher speeds than other modes of transportation (e.g., automobiles, rail, or walking).

(3) While airline travel has increased the mobility of humans, better travel options are still desired. For example, a commercial flight may effectively fly a human thousands of miles, but the places where aircrafts can land is relatively limited. Indeed, not all travel destinations have a commercial airport, and even in towns that have an airport, the airport may not be located particularly close to a person's ultimate destination. Additionally, conventional air travel is typically limited to long-distance travel because air travel is not efficient or feasible for shorter distance travel.

(4) In view of the above, there is a continuing, ongoing need for improved air travel systems and methods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1-4 illustrate a safety and stability device for an aircraft, according to an exemplary embodiment;

(2) FIGS. 5-6 illustrate internal mechanisms for the safety and stability device, according to an exemplary embodiment;

(3) FIG. 7 illustrates an aircraft implementing the safety and stability device, according to an exemplary embodiment; and

(4) FIGS. 8 and 9 illustrate components for connecting a fuselage and other compartments to the safety and stability device, according to an exemplary embodiment.

DETAILED DESCRIPTION

(5) While this invention is susceptible of an embodiment in many different forms, there are shown in the drawings and will be described herein in detail specific embodiments thereof with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention. It is not intended to limit the invention to the specific illustrated embodiments.

(6) Embodiments disclosed herein can include a safety and stability device used in aerospace or aviation. More particularly, the embodiments disclosed herein can include a safety and stability device rotating around a fuselage of an aircraft. The safety and stability device can rotate substantially horizontal to the ground while the aircraft is airborne. The safety and stability device can spin at a particular rate of speed such that the safety and stability device can operate as a gyroscope keeping the aircraft steady and level. Moreover, the safety and stability device can spin for aerodynamic purposes to cut through the air and decrease forward drag on the aircraft fuselage as the aircraft accelerates forward. In addition, the safety and stability device can protect the fuselage and cockpit from damage, such as if the aircraft were to hit a structure, such as buildings, trees, bridges, poles, or any other obstacle.

(7) Due to the benefits described above, the safety and stability device described herein can allow for a personal aircraft or drone to be flown safely by more operators. The safety and stability device increases safety and also provides flight stability that will decrease the learning curve necessary for one to become an effective operator of the aircraft having the safety and stability device described herein. In one embodiment, a drone having the safety and stability device described herein can fly more safely and more stable than conventional drones. In another embodiment described herein, a personal aircraft having the safety and stability device described herein can allow for more air travel, even across shorter distances where conventional air travel would not have been efficient or

feasible.

(8) FIG. 1 illustrates a safety and stability device **100** according to an exemplary embodiment. The safety and stability device **100** can include an inner ring **12** and an outer ring **10** that rotates relative to the inner ring **12**. The inner ring **12** may operate as a hub, and the inner ring **12** may connect to the outer ring **14** via a plurality of bearings **24, 26, 28**, which can assist with the rotation of the outer ring **14**. The safety and stability device **100** can include multiple bearings to meet redundancy requirements for flight set by aviation regulatory bodies, such as the Federal Aviation Administration (FAA). In the embodiment shown in FIGS. 1-6, the safety and stability device **100** can include three bearings, but more bearings can be included. Additionally, a safety and stability device **100** having a single bearing or two bearings is envisioned, should aviation regulations change. The plurality of bearings **24, 26, 28** may be ball, roller or plain bearings, but in a preferred embodiment, the bearings are plain bearings. In any embodiment, the plurality of bearings **24, 26, 28** can decrease the friction between the outer ring **14** and the inner ring **12** when the outer ring **14** rotates relative to the inner ring **12**. The inner ring **12** may include a spool with two grooves, and the bearings **24, 26, 28** and the outer ring **14** may ride within the groove.

(9) The outer ring **14** can connect to or include a geared ring **22**, as shown in FIG. 2. The geared ring **22** can include teeth that can accept and engage with another gear. The geared ring **22** can drive the rotation of the outer ring **14** when another gear transmits rotational movement via torque and speed to the geared ring **22**. Because the outer ring **14** connects to the geared ring **22**, rotational movement of the geared ring **22** also causes rotational movement of the outer ring **14**.

(10) The inner ring **12** can further comprise a dual flange **16** formed on each side of the inner ring **12**. The dual flange **16** can include holes for receiving and holding an axle **20**. The axle **20** can extend across the dual flange **16** such that the axle **20** is held within each hole of the dual flange **16**. In some embodiments, the axle **20** can act as a shaft for a motor **18**. The motor **18** can extend through a hole **14** in the inner ring **12**, and the motor **18** can engage the geared ring **22** through the hole **14**. The motor **18** also can include gear teeth that engage and mesh with the geared ring **22**. As a result, when the motor **18** turns, the gears of the motor **18** may engage with the geared ring **22**, thereby causing the outer ring **10** to turn. The motor **18** can drive the safety and stability device **100** to very fast speeds and a high rotation per minute (RPM). With enough rotational speed, the safety and stability device **100** can generate gyroscopic stability for the safety and stability device **100** and any aircraft component formed inside the inner ring **12**, such as a drone or a personal aircraft. Additionally, the safety and stability device **100** can act as a wing for the aircraft when rotated at a given speed.

(11) The rotation speed of the safety and stability device **100** can vary depending on the translational speed of the aircraft. For example, at higher translational speeds, the rotational speed of the safety and stability device **100** can increase, whereas at lower translational speeds, the rotational speed of the safety and stability device **100** can decrease. The increased rotational speed at higher translational speed encourages stability of the aircraft. In some embodiments, the rotational speed of the safety and stability device **100** can have a direct relationship with the aircraft's translational speed. In another embodiment, the rotational speed of the safety and stability device **100** can have an exponential relationship with the aircraft's translational speed.

(12) More accurately, the rotational speed of the safety and stability device **100** can vary depending on sensor readings that assist in keeping the aircraft fuselage straight and stable during flight. For example, the aircraft may include one or multiple gyroscope sensors to determine whether the aircraft is stable during flight. A processor can receive measurements from the gyroscope sensors, and the processor can adjust the rotational speed of the safety and security device **100** to balance the aircraft fuselage. It should be noted that rotating the safety and security device **100** changes the weight ratio of the entire aircraft. Those having skill in the art will know that flight requires a balance of both lift and aircraft weight as well as balancing thrust and drag. Because changes in rotational speed of the safety and security device **100** can change the weight ratio, then the amount

of lift required for flight also changes and the aircraft may stabilize in view of the change in rotational speed by the safety and security device **100**. The processor is programmed with various formulas and software to control the rotational speed of the safety and security device **100** in response to gyroscope sensor readings, thereby stabilizing the aircraft for flight.

(13) Additionally, those skilled in the art will recognize that aircrafts having spinning objects for flight have “left-turning tendency”, and the processor is further programmed to combat this known phenomenon. In any embodiment, the processor can control the rotational speed of the safety and stability device **100**, and the safety and stability device **100** may comprise the processor. The processor can receive or measure the translational speed of the aircraft, and the processor can apply any formulas by referencing the formula or other relationship programmed into computer-readable memory, and the processor can further send signals to the motor **18** to increase or decrease the rotational speed of the safety and stability device **100** based on the gyroscope readings.

(14) Rotating the safety and security device **100** can have additional benefits other than stability and safety. During translational movement, the nose of an airplane typically increases in heat due to an increase in translational speed due to air drag. The same would be true of the aircrafts disclosed herein, but because the forward most point of the aircrafts disclosed herein is the safety and security device **100**, which rotates, the rotation of the safety and security device **100** can dissipate the generated heat across the entire safety and security device **100**. This dissipation can decrease the heat load on the aircraft and also decrease the need for heat plates at a nose or front tip of an aircraft.

(15) During takeoff, the safety and security device **100** can apply disk loading principals, by slowing ramping up the rotational movement of the safety and security device **100** to a stable rotational speed (e.g., 2500 RPM). The rotation of the safety and security device **100** can provide some lift, but not enough for takeoff or extended flight, so the aircrafts disclosed herein may have additional propellers or the like to provide extended flight. Additional propellers are particularly necessary for takeoff, which lifts dead weight even though the weight ratio of the aircraft changes due to the rotation of the safety and security device **100**.

(16) FIG. 3 illustrates the safety and stability device **100** from a side perspective, and FIG. 4 illustrates the safety and stability device from a top perspective. In some embodiments, the outer ring **10** may comprise a soft material such as rubber and may be filled with air, similar to a tire. Alternatively, the outer ring **10** may comprise any material that softens impact of the outer ring **14** with any solid structure that it strikes, and such material may comprise an air-filled polymer, polyethylene foam, gels or any other soft material that reduces the impact of any forceful strike with a solid object.

(17) FIG. 5 illustrates the safety and stability device **100** at the cross section illustrated by line 5-5 in FIG. 1, and FIG. 6 illustrates the safety and stability device **100** at the cross section illustrated by line 6-6 in FIG. 5. FIGS. 5 and 6 better illustrate the interaction between the geared motor **18** and the geared ring **22** that causes rotation of the outer ring **14**.

(18) Importantly, the safety and stability device **100** rotates in a substantially horizontal plane. Because the safety and stability device **100** rotates horizontally, the safety and stability device **100** provides gyroscopic stability that stabilizes the safety and stability device **100** and anything connected to the safety and stability device **100** within the circular area created by the inner ring **12**. Moreover, the horizontal rotation of the safety and stability device **100** provides protection of the aircraft fuselage at all sides of the aircraft. That is, the safety and stability device can protect the aircraft should the aircraft strike any structure during translational movement.

(19) Notably, the safety and stability device **100** lacks a hub within the circular inner area created by the inner ring **12**. The safety and stability device **100** lacks a hub so that additional aviation equipment may exist within and connect to the safety and stability device **100**. The additional aviation equipment can include any aviation equipment including drone equipment, a cockpit, a fuselage, a passenger cabin, a cargo compartment, jet engines, flaps, motors, propellers, canards, or

any other equipment used in aviation.

(20) FIG. 7 illustrates an aircraft embodiment **700** implementing the safety and stability device **100** described above with reference to FIGS. 1-6. As shown in FIG. 7, the aircraft **702** fits within the hole created by the inner ring **12** (because the safety and stability device **100** lacks a hub). Moreover, the outer ring **14** rotates around the aircraft **702**. In one embodiment, the aircraft **702** can include multiple propellers **710** all formed within the inner ring **12** of the safety and stability device **100**. The number of propellers **710** can vary, with FIG. 7 illustrating a six-propeller embodiment, but other embodiments may have a single propeller, two propellers, four propellers, or any number of propellers. Each propeller may be associated with a respective motor driving the propeller. Additionally, the propellers **710** may increase rotation speed to increase the altitude of the drone embodiment **700**, and the propellers **710** may decrease rotation speed to decrease the altitude of the drone embodiment **700**. Also, in an embodiment, where the propellers **710** are used to turn the drone embodiment **700**, the clockwise turning propellers may decrease in speed while the counterclockwise spinning propellers increase in speed **700** to turn the drone **700** to the left, whereas the counterclockwise propellers may decrease in speed while the clockwise propellers increase in speed **700** to turn the drone **700** to the right. In another embodiment, flaps on the wings or tailfin in conjunction with aircraft tilt may cause the drone embodiment **700** to turn left or right. In some embodiments, the aircraft **702** can comprise a drone, while in another embodiment, the aircraft **702** can comprise a human-piloted aircraft or an aircraft that carries and flies humans as a mode of transportation.

(21) The aircraft **702** can further include a top disc **720** and a bottom disk (not shown), where the top and bottom disks **720** can have a radius substantially similar to the radius of the inner ring **12**. In this way, the aircraft **702** can connect to the inner ring **12** of the safety and stability device **100**. In some embodiments, the top and bottom disks **720** can be welded to the inner ring **12** or together form the inner ring **12**, but other connection methods are contemplated, such as a removable and replaceable option to swap out one aircraft **702** for another. The propellers **710** can spin between the top and bottom disks **720**, and the top and bottom disks **720** may include grills placed above and below each propeller **710**.

(22) Referring again to FIG. 7, the aircraft **702** can include a canard **730** acting as additional wings for flight control of the drone embodiment **700**. As shown, the canard **730** can include a combination of front and rear wings, as is well-understood in the art of aviation. The canard **730** can connect to and extend outward from a main body **735**, which can house important internal components, such as a fuel tank, a battery, a main computer, propeller motors, wireless connection with a control unit operated by a human to fly the drone, or any other internal component, as would be well-understood in the field of drone aviation.

(23) In the embodiment shown in FIG. 7, flight can be controlled by the combination of the plurality of propellers **710**, the canard **730**, the safety and stability device **100**, and propulsion engines **740**. In this embodiment, the propulsion engines **740** can generate translational or forward movement of the aircraft, the plurality of propellers **710** can primarily provide the upward lift of the drone embodiment **700**, the canard **730** can provide lift, control, and stability for the drone embodiment **700**, and the safety and stability device **100** can provide stability, some lift, and aerodynamic benefits by cutting through the air at a high rate of speed, thereby decreasing the drag on the aircraft **702**. In some embodiments, the propellers **710** may provide lift only until the canard **730** and the safety and stability device **100** are able to maintain lift alone, which may occur at high translational speeds resulting from propulsion provided by the propulsion engine **740**, at which time the propellers may stop spinning at a high rate of speed. In some embodiments, the propulsion engines **740** may be omitted in favor of a drone embodiment **700** where the propellers **710** provide translational movement and lift. In yet another embodiment, the canard **730** and/or the propulsion engines **740** may be omitted. In other words, any drone configuration capable of connection to the safety and stability device **100** is envisioned.

(24) Referring now to FIGS. 8-9, FIGS. 8-9 illustrate a personal aircraft embodiment **800** configured for use with the safety and stability device **100**. More particularly, FIGS. 8-9 illustrate components for connecting a fuselage and other compartments to the safety and stability device **100**. In some embodiments, the personal aircraft embodiment **800** may operate similar to a personal helicopter for providing relatively short distance air travel. In this way, the personal aircraft embodiment **800** may include a main propeller and tail rotor for control, turning, and reactional torque to prevent spinning of the personal aircraft embodiment **800**. The main propeller and rear rotor are not illustrated, as those features would be well-understood by those of skill in the art.

(25) FIG. 8 illustrates a first personal aircraft embodiment **800A**, where the safety and stability device **100** connects to a fuselage **802** via an upper rib **804** and a lower rib **806**. FIG. 8 illustrates a cross-section of one upper rib **804** and one lower rib **806**, but the first personal aircraft embodiment **800A** may include multiple upper ribs **804** and multiple lower ribs **806** placed evenly around the safety and stability device **100**. That is, the ribs **804**, **806** may surround or encircle the fuselage **802** and connect to multiple points of the circular safety and security device **100**. As shown in FIG. 8, the upper rib **804** and the lower rib **806** connect to the inner ring **12** and also surround the plurality of bearings **24**, **26**, **28**, and the upper rib **804** and the lower rib **806** can also connect to the fuselage **802** and a lower cargo bay **810**. In some embodiments, the fuselage may comprise a cockpit of the personal aircraft embodiment **800** where a human operator may sit and fly the personal aircraft embodiment **800**. The upper rib **804** and the lower rib **806** may include holes for aerodynamic purposes to reduce drag during translational movement of the first personal aircraft embodiment **800A**.

(26) FIG. 9 illustrates a second personal aircraft embodiment **800B**, where the safety and stability device **100** connects to a fuselage **802** via brackets **910**. FIG. 9 illustrates a cross-section of one bracket **910**, but the second personal aircraft embodiment **800B** may include multiple brackets **910** placed evenly around the safety and stability device **100**. That is, the brackets **910** may surround or encircle the fuselage **802** and connect to multiple points of the circular safety and security device **100**. As shown in FIG. 9, the bracket **910** connects to the inner ring **12** and the fuselage **802** and a lower cargo bay **810**. The bracket **910** may include a hole for aerodynamic purposes to reduce drag during translational movement of the personal aircraft embodiment, but the hole may further include cross bracing to increase the strength of the bracket **910**. Additionally, the bracket **910** can include one or more L-brackets **912** that connect the inner ring **12** of the safety and stability device **100** to the bracket **910**. While L-Brackets are shown in the embodiment shown in FIG. 9, any other connection mechanism are contemplated.

(27) In either embodiment illustrated in FIGS. 8-9, the fuselage **802** may disconnect easily from the brackets **910** or the upper and lower ribs **804**, **806** in the event of an emergency. By disconnecting the fuselage **802**, the fuselage **802** may eject or fall from the rest of the aircraft, and the fuselage **802** may include a parachute to allow a safe landing of the fuselage **802**.

(28) Although the embodiments described in FIGS. 8-9 describe a personal aircraft, the embodiments described herein can increase in size by increasing the circumference of the safety and stability device **100** to allow for additional passengers. The exemplary embodiments described herein are not limited to a single seater aircraft.

(29) As has been shown, the exemplary embodiments described herein illustrate an aircraft having a safety and stability device that surrounds an aircraft's fuselage for protection of the aircraft and stability of the aircraft. An aircraft having a rotating wing can provide increased stability and protection, which may lower the learning curve necessary for more people to use air travel to travel shorter distances typically only available by automobile, bicycle or other short-distance travel options. Therefore, the aircraft having the rotational wing described herein represents a dramatic improvement over the prior art.

(30) Although a few embodiments have been described in detail above, other modifications are possible. For example, other components may be added to or removed from the described systems,

and other embodiments may be within the scope of the invention.

(31) From the foregoing, it will be observed that numerous variations and modifications may be effected without departing from the spirit and scope of the invention. It is to be understood that no limitation with respect to the specific system or method described herein is intended or should be inferred. It is, of course, intended to cover all such modifications as fall within the spirit and scope of the invention.

Claims

1. An aircraft comprising: a safety and stability device comprising: an inner ring forming a circular inner area; an outer ring that rotates relative to the inner ring; and a motor connected to the inner ring that drives rotation of the outer ring relative to the inner ring, wherein the safety and stability device rotates in a substantially horizontal plane and at a rotational speed sufficient to provide gyroscopic stability for the aircraft; a fuselage connected to the inner ring formed within the circular inner area; and at least one propeller connected to the fuselage and formed within the circular inner area, where the outer ring surrounds the inner ring and the fuselage, and wherein the outer ring rotates during flight.
2. The aircraft of claim 1, wherein the fuselage further comprises a cockpit.
3. The aircraft of claim 1, wherein the fuselage comprises a main body and a canard.
4. The aircraft of claim 3 further comprising a top plate connected to the inner ring and a bottom plate connected to an inner ring, wherein aviation equipment is formed between the top plate and the bottom plate.
5. The aircraft of claim 4 wherein the aviation equipment at least comprises the at least one propeller.
6. The aircraft of claim 1 further comprising at least one upper rib and a lower rib, wherein the upper rib is connected to the inner ring and the fuselage, and wherein the lower rib is connected to the inner ring and a cargo bay.
7. The aircraft of claim 1 further comprising at least one bracket, wherein the bracket is connected to the inner ring and the fuselage.
8. The aircraft of claim 1 further comprising propulsion engines to generate translational movement of the aircraft.
9. The aircraft of claim 1 further comprising a plurality of bearings existing between the inner ring and the outer ring to decrease friction when the outer ring rotates relative to the inner ring.
10. The aircraft of claim 9 wherein each of the plurality of bearings comprises a ball, roller, or plain bearing.
11. The aircraft of claim 1 further comprising a geared ring connected to the outer ring.
12. The aircraft of claim 11 wherein the motor further comprises a gear that engages the geared ring and transmits rotational movement to the geared ring and thereby the outer ring.
13. The aircraft of claim 12 wherein the gear engages the geared ring through a hole formed in the inner ring.
14. The aircraft of claim 12 wherein the inner ring further comprises a pair of flanges holding an axle, and wherein the axle holds the gear and allows for rotational movement of the gear.
15. The aircraft of claim 1 wherein the outer wheel comprises a soft material for absorbing any impact made with a structure, the soft material comprising an air-filled polymer, polyethylene foam, or gel.
16. The aircraft of claim 1 wherein the rotational speed sufficient to provide gyroscopic stability for the aircraft depends on a translational speed of the aircraft, and wherein the rotational speed sufficient to provide gyroscopic stability for the aircraft increases with an increase in the translational speed of the aircraft.
17. An aircraft comprising: a safety and stability device comprising: an inner ring forming a

circular inner area; an outer ring that rotates relative to the inner ring; and a motor connected to the inner ring that drives rotation of the outer ring relative to the inner ring, wherein the safety and stability device rotates in a substantially horizontal plane; a fuselage connected to the inner ring; and at least one propeller formed within the circular inner area, where the outer ring surrounds the inner ring and the fuselage, and wherein the outer ring rotates during flight.
